# Transforming derivatives

We already know from our table of Laplace transforms that we can transform first and second derivative functions.

$$\mathscr{L}(f'(t)) = sF(s) - f(0)$$

$$\mathscr{L}(f''(t)) = s^2F(s) - sf(0) - f'(0)$$

In this lesson we want to understand how we arrived at these formulas. We'll also be using these in an upcoming lesson in order to solve initial value problems.

## Transform of a first derivative

We can start with a simple substitution into the definition of the Laplace transform, replacing $f(t)$ in that definition with its derivative $f'(t)$.

$$\mathscr{L}(f'(t)) = \int_0^\infty e^{-st}f'(t)\ dt$$

But we'd still prefer to have an expression in terms of $f(t)$, instead of $f'(t)$, since we're usually starting with $f(t)$. Luckily, we can put this equation in terms of $f(t)$ by applying integration by parts to the integral.

We'll differentiate the $e^{-st}$ part and integrate the $f'(t)$ part, which means we'll set up our parts as

$$u = e^{-st} \qquad dv = f'(t)\ dt$$

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$$du = -se^{-st}\,dt \qquad\qquad v = f(t)$$

Then integration by parts gives us

$$\mathscr{L}(f'(t)) = uv\Big|_0^\infty - \int_0^\infty v\,du$$

$$\mathscr{L}(f'(t)) = (e^{-st})(f(t))\Big|_0^\infty - \int_0^\infty f(t)(-se^{-st})\,dt$$

$$\mathscr{L}(f'(t)) = e^{-st}f(t)\Big|_0^\infty + s\int_0^\infty e^{-st}f(t)\,dt$$

Now assuming $f(t)$ is of exponential type, then $e^{-st}f(t)$ converges to $0$, and we can simplify this expression for the Laplace transform of the derivative as

$$\mathscr{L}(f'(t)) = \lim_{t\to\infty} e^{-st}f(t) - e^{-s(0)}f(0) + s\int_0^\infty e^{-st}f(t)\,dt$$

$$\mathscr{L}(f'(t)) = 0 - (1)f(0) + s\int_0^\infty e^{-st}f(t)\,dt$$

$$\mathscr{L}(f'(t)) = -f(0) + s\int_0^\infty e^{-st}f(t)\,dt$$

Furthermore, we see that the integral that remains is simply the definition of the Laplace transform,

$$F(s) = \int_0^\infty e^{-st}f(t)\,dt$$

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which means we can substitute and replace it with $F(s)$, in order to get a clean and concise definition for the **Laplace transform of a first derivative**.

$$\mathscr{L}(f'(t)) = -f(0) + sF(s)$$

$$\mathscr{L}(f'(t)) = sF(s) - f(0)$$

## Transform of a second derivative

Using the same method, we can extend this process to find the **Laplace transform of a second derivative**, which is given as

$$\mathscr{L}(f''(t)) = s^2F(s) - sf(0) - f'(0)$$

Notice that the transform of the first derivative requires us to know $f(0)$, and the transform of the second derivative requires us to know both $f(0)$ and $f'(0)$.

In other words, we won't be able to take transforms of these kinds of derivative functions unless we have the starting points for the original function $f(t)$ and its derivative $f'(t)$.

Let's do an example so that we can practice with these derivative formulas.

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**Example**

Find the Laplace transforms of $y'(x)$ and $y''(x)$, given $y(0) = 1$ and $y'(0) = 0$.

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We'll start by rewriting the formula for the Laplace transform of a first derivative by replacing the function $f'(t)$ with the derivative function we were given, $y'(x)$. This will also change the transform $F$ to the transform $Y$, and the initial condition from $f(0)$ to $y(0)$.

$$\mathscr{L}(f'(t)) = sF(s) - f(0)$$

$$\mathscr{L}(y'(x)) = sY(s) - y(0)$$

Now we can make a substitution into the transform equation for the initial condition.

$$\mathscr{L}(y'(x)) = sY(s) - 1$$

Now we'll rewrite the formula for the Laplace transform of a second derivative so that it's in terms of $y$ and $x$ instead of $y$ and $t$.

$$\mathscr{L}(f''(t)) = s^2F(s) - sf(0) - f'(0)$$

$$\mathscr{L}(y''(x)) = s^2Y(s) - sy(0) - y'(0)$$

Then we'll substitute for the initial conditions.

$$\mathscr{L}(y''(x)) = s^2Y(s) - s(1) - 0$$

$$\mathscr{L}(y''(x)) = s^2Y(s) - s$$

So for any function $y(x)$ with the initial conditions $y(0) = 1$ and $y'(0) = 0$, the Laplace transforms of its first and second derivatives will be

$$\mathscr{L}(y'(x)) = sY(s) - 1$$

$$\mathscr{L}(y''(x)) = s^2Y(s) - s$$

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